Software Engineering

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

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Section : H
Project : Customizable Subscription Box Scheduler

Epic 1: Subscription Management

Description:

Enable subscribers to create a new subscription and manage its lifecycle, including pausing upcoming deliveries and skipping a billing cycle before the monthly renewal cutoff.

User Stories:

- SP-4 – Create Subscription — 2 points
- SP-5 – Pause Subscription — 2 points
- SP-6 – Skip Billing Cycle — 2 points

Total Story Points: 6

Epic 2: Monthly Box Customization

Description:

Allow subscribers to view the current monthly box, swap items using preference-based options, and save their updated selection before the editing window closes.

User Stories:

- SP-7 – View Monthly Box Items — 1 point
- SP-8 – Swap Monthly Box Items — 3 points
- SP-9 – Save Updated Selection — 3 points

Total Story Points: 7

Epic 3: Fulfillment Processing

Description:

Provide the Fulfillment Lead with the ability to view finalized subscriber selections, generate the monthly fulfillment manifest, and export shipping labels for the batch.

User Stories:

- SP-10 – View Finalized Selections — 3 points
- SP-11 – Generate Fulfillment Manifest — 5 points
- SP-12 – Export Shipping Labels — 5 points

Total Story Points: 13

Sprint Execution

A one-week Sprint 1 was created in Jira with all nine user stories from the three epics — Subscription Management, Monthly Box Customization, and Fulfillment Processing — added to the sprint scope, totalling 26 story points. The sprint was started and stories were progressed through the Agile workflow from To Do → In Progress → Done. By the end of Sprint 1, five stories were completed — Create Subscription, Swap Monthly Box Items, Save Updated Selection, Generate Fulfillment Manifest, and Export Shipping Labels — accounting for 18 completed story points. The remaining four stories — Pause Subscription, Skip Billing Cycle, View Monthly Box Items, and View Finalized Selections (8 story points) — were carried into a second one-week sprint. Sprint 2 was used to complete this remaining subscription-management and box-customization functionality, and all four stories were moved to Done by the end of the sprint. Across both sprints, all 9 planned stories were completed, covering the full 26 story points.

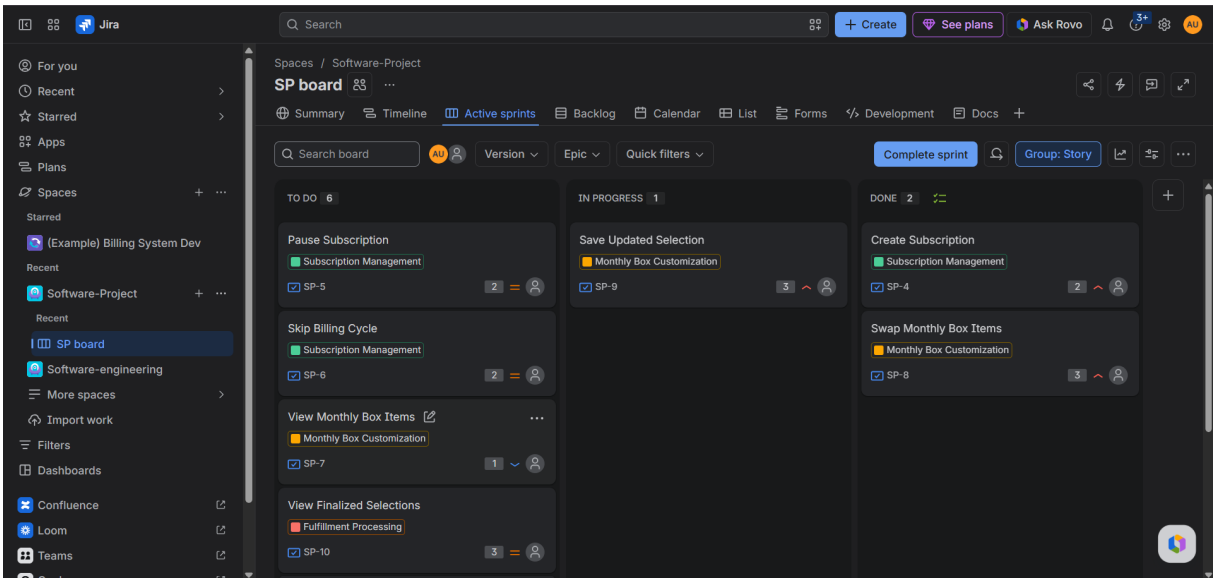
Story Point Estimation

The lab requires assigning story points using the Fibonacci sequence. Story points were estimated based on the relative effort, complexity, and uncertainty associated with each user story.

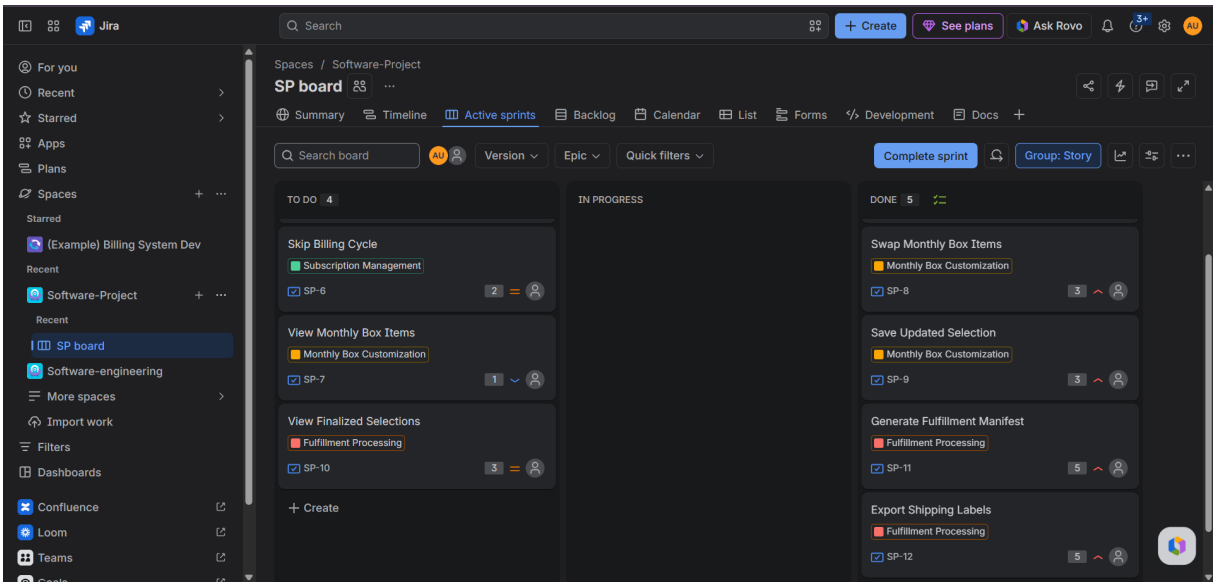
Story Points :

Story	Story Points
SP-4 – Create Subscription	2
SP-5 – Pause Subscription	2
SP-6 – Skip Billing Cycle	2
SP-7 – View Monthly Box Items	1
SP-8 – Swap Monthly Box Items	3
SP-9 – Save Updated Selection	3
SP-10 – View Finalized Selections	3
SP-11 – Generate Fulfillment Manifest	5
SP-12 – Export Shipping Labels	5
Total	**26**

JIRA Sprint 1 Completion Screenshot :

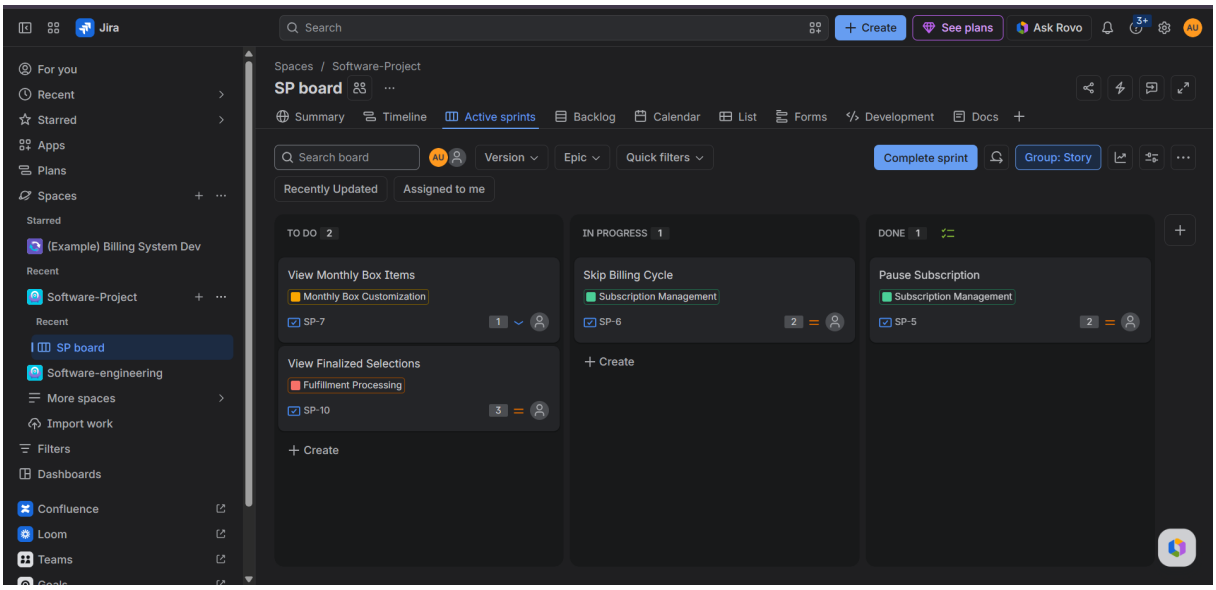


Sprint 1 in progress — Create Subscription and Swap Monthly Box Items done, Save Updated Selection in progress.

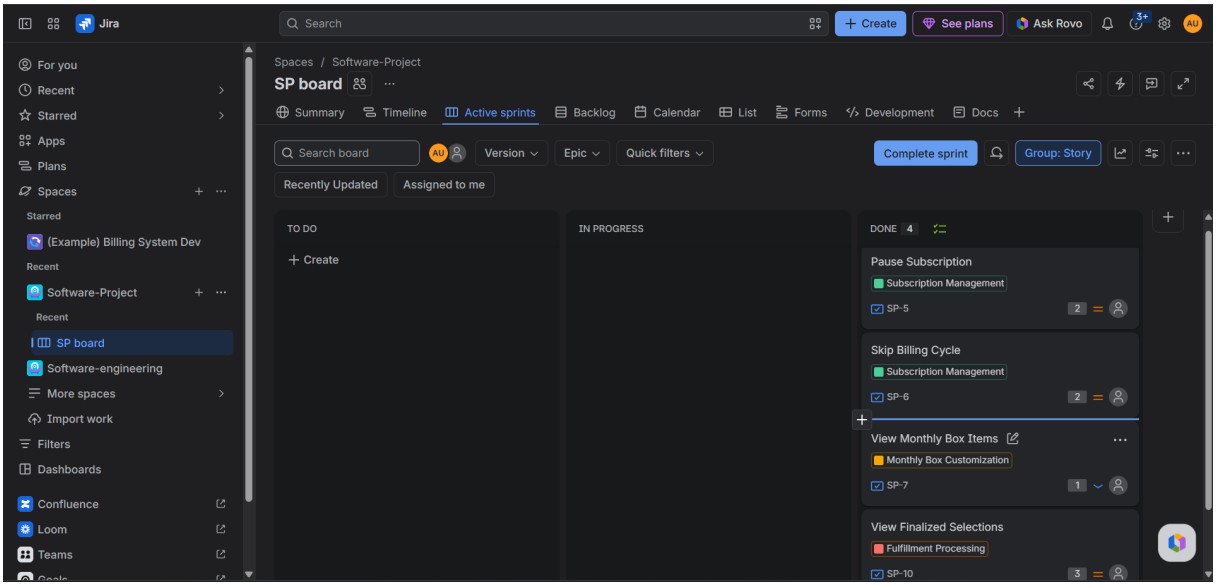


Sprint 1 near completion — 5 of 9 stories moved to Done (18 story points); the remaining 4 stories were carried into Sprint 2.

JIRA Sprint 2 Completion Screenshot :

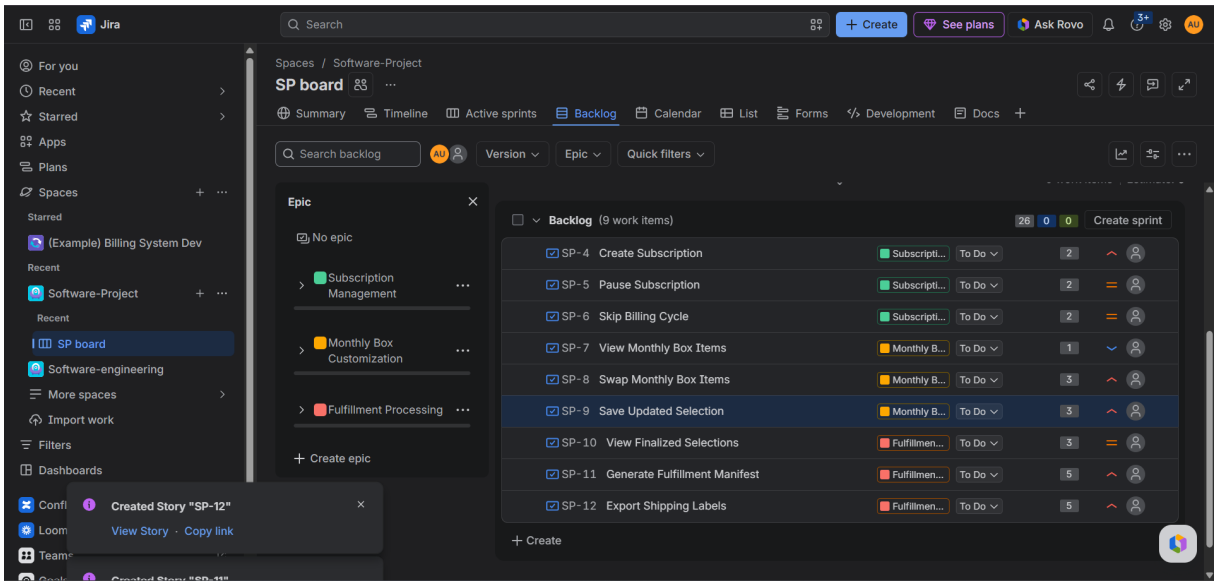


Sprint 2 in progress — Pause Subscription done, Skip Billing Cycle in progress.



Sprint 2 completed — all 4 remaining stories (8 story points) moved to Done.

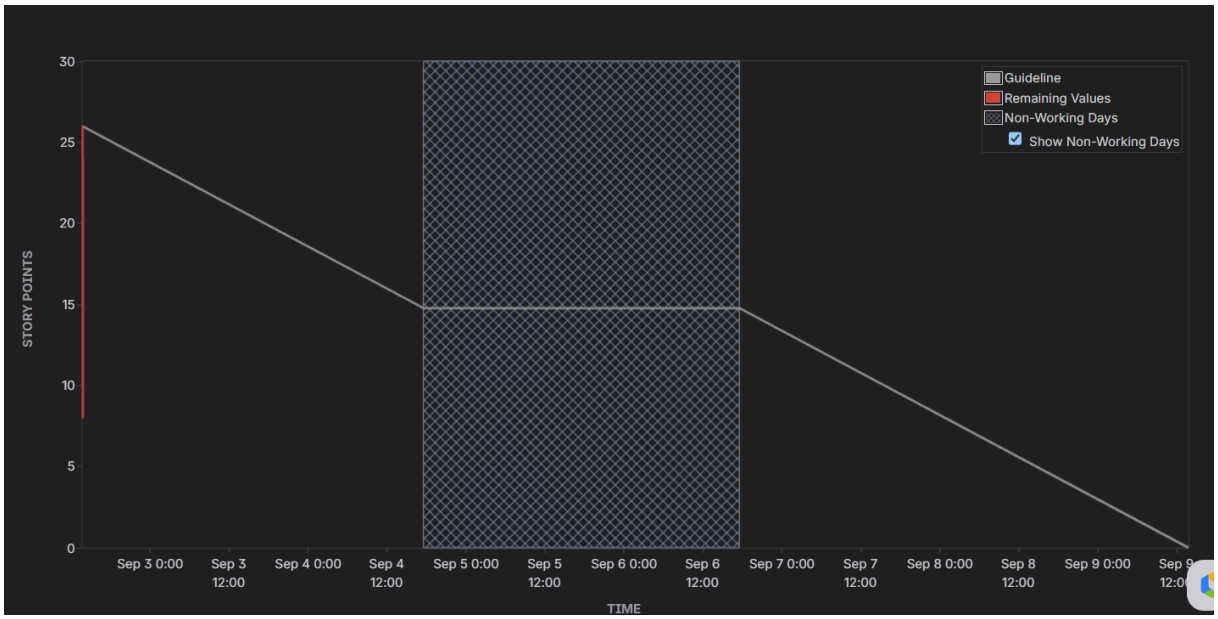
JIRA All Epics and Stories with Story Points :



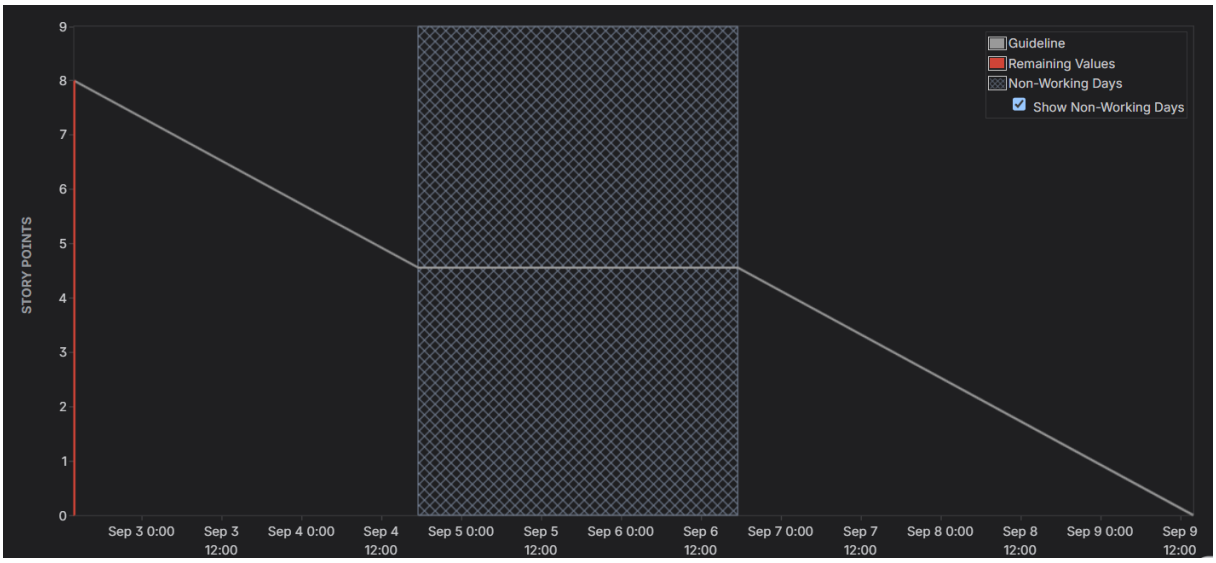
SP board backlog showing all 9 user stories across the 3 epics — Subscription Management, Monthly Box Customization, and Fulfillment Processing — with their assigned story points (total 26).

Burndown Chart :

SP Sprint 1 :-



SP Sprint 2 :-



Reflection Questions :

1. Did your estimations reflect the actual effort?

Ans : Yes, the story point estimations gave a reasonable representation of the effort involved in each user story. Fulfillment-side tasks such as Generate Fulfillment Manifest and Export Shipping Labels were assigned higher points (5) because they involve batch processing and integration with the shipping workflow. Simpler, well-understood actions such as Create Subscription, Pause Subscription, and Skip Billing Cycle needed fewer points (2), and viewing the current monthly box (View Monthly Box Items) required the least effort (1). The Fibonacci scale made the increasing complexity and uncertainty between these tasks easy to represent.

2. Was your backlog well-prioritized?

Ans : Yes, the backlog was prioritized according to the core purpose of the Customizable Subscription Box Scheduler. Subscription lifecycle management and monthly box customization were treated as the primary subscriber-facing functionality, since these directly implement the core use case of customizing and finalizing a monthly box before the renewal cutoff. Fulfillment processing — viewing finalized selections, generating the manifest, and exporting shipping labels — was prioritized next, as it depends on the subscriber-side selections being finalized first.

3. How did your simulated sprint align with your plan?

Ans : The simulated sprints aligned well with the planned backlog. All nine stories were pulled into Sprint 1; five of them — covering subscription creation and the full box customization and fulfillment path — were completed within the sprint (18 story points). The remaining four stories, covering pausing, skipping, viewing the monthly box, and viewing finalized selections (8 story points), were carried over and completed in Sprint 2. Both sprints moved their stories through To Do → In Progress → Done, and all 9 stories across both epics' worth of functionality were completed by the end of Sprint 2.

4. What insights did the burndown chart give about your team's capacity?

Ans : The Sprint 1 burndown chart (scaled to 26 story points) showed a sharp initial drop as the first stories were completed, followed by a plateau while the remaining four stories were still in progress at the end of the time-box — indicating the sprint scope was slightly optimistic for a single one-week iteration. The Sprint 2 burndown chart, scaled to the smaller remaining scope of 8 story points, showed a more consistent burn down to zero, suggesting the reduced, more realistic scope was a better match for the team's actual capacity in a one-week sprint.